H M Muntakim Rahman

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LinkedIn GitHub Portfolio
Chittagong, Bangladesh

Professional Profile

Computer Science and Engineering graduate from CUET who builds, evaluates and deploys LLM systems, including a bilingual English–Bangla multi-agent legal assistant with retrieval-augmented generation, live as a web app. Found and fixed a data leak that inflated its accuracy by 23 points. Interested in AI engineering and applied LLM research.

Education

Chittagong University of Engineering & Technology (CUET) March 2022 – July 2026
Bachelor of Science in Computer Science & Engineering *CGPA: 3.56/4.00* *Chittagong, Bangladesh*

Chittagong College 2018 – 2020
Higher Secondary Certificate (HSC) *GPA: 5.00/5.00* *Chittagong, Bangladesh*

Chattogram Cantonment Public College 2018
Secondary School Certificate (SSC) *GPA: 5.00/5.00* *Chittagong, Bangladesh*

Research Experience

BanglaLex: A Multi-Agent AI Framework for Legal Assistance in Bangladesh GitHub
Undergraduate Thesis, CUET *Supervisor: Md. Al-Mamun Provath, Assistant Professor* *2025 – 2026*

- Built a four-stage LLM pipeline in Python that turns a plain-language complaint in English or Bangla into a predicted outcome with a bilingual explanation, grounded in retrieved statutes and similar past cases; runs on GPT-OSS-120B (Groq) or Gemini 3.1 Flash-Lite without code changes.
- Built a dataset of 164 Supreme Court cases (land, contract, service) labelled with outcomes (Fleiss' $\kappa = 0.807$ on a blind relabelling by three law students), and a FAISS index of 43,015 passages from 1,281 active statutes (15–17 ms per query).
- Reached 70.0% and 75.0% accuracy on 60 held-out real cases against a 61.7% majority-class floor, answering every case where the bare LLMs declined 17–22% of them.
- Found and fixed an evaluation leak in which LLM-paraphrased copies of test cases entered their own precedent pool, inflating accuracy by 23 points; reported it as a finding in the paper.
- Paper submitted to ICCIT 2026 (under review).

Technical Projects

BanglaLex Legal Assistant Dashboard GitHub Live Demo
Python, Streamlit, FAISS, Sentence-Transformers, Groq API, Gemini API

- Built and deployed an interactive Streamlit app where users describe a land, contract or service dispute in English or Bangla and receive a predicted outcome, applicable statutes and next steps in the same language.
- Added a live progress tracker across the four pipeline stages, a Groq/Gemini backbone switch, one-click examples in both languages, and an expandable view of the reasoning with supporting and risk factors.

GafferBot: Fantasy Premier League Planning Agent GitHub
Python, pandas, SciPy, PuLP/HiGHS, GitHub Actions, Telegram Bot API

- Built an expected-points model for every Premier League player from minutes, xG/xA, opponent strength and injury news, with Poisson estimates for clean sheets, goals conceded and saves.
- Formulated transfers, starting XI and captaincy as a mixed-integer program over a five-gameweek horizon under all FPL rules (budget, squad quotas, club limits, formations).
- In a walk-forward backtest over 34 gameweeks of 2025/26, the model's top-11 picks scored 54.6 points per gameweek against 47.6 for a season-average baseline (Spearman 0.750 vs 0.691).
- Runs on a GitHub Actions schedule and sends the plan to Telegram before each deadline.

WayFinder: Public Transport Route and Fare Finder
HTML, CSS, JavaScript, Google Maps API, Node.js, Express.js, MongoDB *Frontend Developer, team of 3*

- Co-developed the frontend of a web app where commuters search a trip, see the route on Google Maps with place autocomplete, and get a fare estimate from an Express/MongoDB backend.
- Built responsive pages for trip search, bus routes, an admin panel and the team page together with one other frontend developer.

Technical Skills

Programming: Python, C, C++, Java, SQL, JavaScript, HTML, CSS
Computer Science: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
LLM & AI Engineering: LLM APIs (Groq, Gemini), Retrieval-Augmented Generation, Multi-Agent Pipelines, Prompt Engineering, Structured JSON Outputs
Machine Learning & Data: PyTorch, Hugging Face Transformers, Sentence-Transformers, FAISS, scikit-learn, pandas, NumPy, SciPy
Optimisation: Mixed-Integer Programming (PuLP, HiGHS), Poisson Modelling
Research Methods: Leakage-Safe Evaluation, Significance Testing (McNemar's Test), Inter-Annotator Agreement (Fleiss' κ), Walk-Forward Backtesting
Deployment: Streamlit, GitHub Actions, Telegram Bot API, Google Maps JavaScript API
Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Experience

RedDot Digital Limited November 2025
Industrial Trainee

- Trained in a 9-member cross-functional team under senior engineering leadership, following the company's development workflow from requirements to delivery.
- Contributed to an AI-integrated product concept through ideation, UI/UX design and user-flow development, and presented it to senior technical and executive stakeholders.

Leadership & Organisational Experience

Public Relations Secretary July 2025 – August 2026
IEEE Computer Society CUET Student Branch Chapter

Publicity Secretary August 2025 – August 2026
Joyoddhoney, CUET

Certifications

Front-End Development Training (80 Hours) — CSE, CUET (EDGE Project) April 2025
Fundamentals of Agents (Unit 1, Agents Course) — Hugging Face April 2026

References

Md. Al-Mamun Provath, Assistant Professor, Department of Computer Science & Engineering, CUET

Dr. A M Muntasir Rahman, Postdoctoral Researcher, Rutgers University, USA

Contact details available on request.